

Customer Behavior & Retail Sales Analytics Report

A structured analysis of ~3,900 customer records and transaction-level purchase data — uncovering how demographics, subscriptions, discounts and shipping choices shape revenue across a retail product catalog.

3.9K

CUSTOMERS ANALYZED

\$233K

TOTAL REVENUE COVERED

4

PRODUCT CATEGORIES

3.75

AVG. REVIEW RATING

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Python (pandas) · SQL querying & data modeling · Dashboard design & visualization

Executive Summary & Approach

This project analyzes a retail customer-transaction dataset to understand who the customers are, how they behave, and which levers — subscription status, discounting, shipping choice, age group — actually move revenue. The workflow moved from Python-based cleaning and feature engineering, through SQL querying and ranking logic, into a single interactive BI dashboard for stakeholder-facing exploration.

WHAT WAS DONE

1 Data cleaning & feature engineering

Used Python (pandas) to profile, clean and enrich the raw 3,900-row dataset — imputing missing ratings, mapping purchase frequency to numeric days, deriving age groups, and dropping a redundant column — before loading it into PostgreSQL. Full detail on page 03.

2 Data querying & aggregation

Wrote SQL queries against customer and purchase tables (839 transaction rows joined to customer-level attributes) to compute revenue, averages, and rankings across multiple dimensions: gender, age group, subscription status, category, and shipping type.

3 Segmentation & ranking logic

Built window-function style rankings (e.g. top items per category, top-rated and most-discounted products) rather than flat totals, to surface relative winners inside each group instead of just the biggest numbers overall.

4 Dashboard consolidation

Brought the query outputs together into one filterable dashboard (subscription status, gender, category, shipping type) with KPI cards, a donut, bar and horizontal-bar charts, so the same numbers could be sliced live instead of read off separate tables.

HEADLINE NUMBERS

3,900

TOTAL CUSTOMERS

\$59.76

AVG. PURCHASE AMOUNT

3,116

LOYAL-SEGMENT CUSTOMERS

Core takeaway: revenue is driven far more by **customer volume** than by any single high-value segment — non-subscribers, standard-shipping and "loyal" segment customers dominate simply because they are the largest groups, while average spend per person is remarkably flat across almost every slice.

From Raw CSV to Warehouse-Ready Data

Before any SQL querying or dashboarding, the raw **customer_shopping_behavior.csv** (3,900 rows, 18 columns) was profiled, cleaned and enriched in Python with pandas, then pushed into a PostgreSQL database — the same table the SQL analysis in this report runs against.

3,900 × 18

RAW ROWS × COLUMNS

37 → 0

NULLS IN REVIEW RATING

18 → 15

COLUMNS AFTER CLEANUP

1 Load & profile

Loaded the CSV with **pandas.read_csv** and profiled every column with **df.describe(include='all')** — confirming 3,900 complete customer records, ages 18–70, purchase amounts \$20–\$100 (mean \$59.76), across 50 locations and 4 seasons.

2 Handle missing values

df.isnull().sum() found exactly one gap: 37 missing **Review Rating** values. Rather than a single global fill, each missing rating was imputed with its own category's median via **groupby('Category')['Review Rating'].transform()** — so a missing Footwear rating is filled from Footwear's median, not the whole catalog's.

3 Engineer purchase_frequency_days

The categorical **frequency_of_purchases** field ("Weekly", "Fortnightly", "Annually"...) was mapped to a numeric day-count (Weekly → 7, Monthly → 30, Annually → 365, etc.) with a dictionary + **.map()**, turning a text label into a number that can be aggregated, ranked and compared.

4 Engineer age_group

pd.qcut(df['age'], q=4, labels=[...]) split customers into four equal-sized age quartiles — Young Adult, Adult, Middle-aged, Senior — the same segments used in the age-based revenue breakdown later in this report.

5 Drop a redundant column

A quick check — **(df['discount_applied'] == df['promo_code_used']).all()** — returned **True** for all 3,900 rows: the two columns were perfect duplicates of each other. **promo_code_used** was dropped rather than carrying a redundant signal into the database.

LOADING INTO POSTGRESQL

```
from sqlalchemy import create_engine
engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")
df.to_sql("customer", engine, if_exists="replace", index=False)
# Data successfully loaded into table 'customer' in database 'customer_behavior'.
```

The cleaned, 15-column DataFrame was written with **SQLAlchemy + psycopg2** into a **customer** table — handing off from Python to the SQL work next.

Who the Customers Are

CUSTOMER SEGMENTS

SEGMENT	CUSTOMERS	SHARE	
Loyal	3,116	79.9%	
Returning	701	18.0%	
New	83	2.1%	

The base is overwhelmingly loyal — four out of five customers already fall into the "Loyal" bucket, and genuinely new customers are a thin 2% sliver. This reads as a mature, retention-heavy customer base rather than one still in an acquisition phase.

REVENUE BY GENDER

GENDER	REVENUE	SHARE
Male	\$157,890	67.7%
Female	\$75,191	32.3%

Male customers generate roughly **2.1x** the revenue of female customers — the largest single skew found anywhere in the dataset, and worth treating as a headline finding rather than a footnote.

SUBSCRIPTION STATUS VS. SPENDING

SUBSCRIBED	CUSTOMERS	AVG. SPEND	TOTAL REVENUE
No	2,847	\$59.87	\$170,436
Yes	1,053	\$59.49	\$62,645

Subscription doesn't change how much someone spends
Average spend is virtually identical whether a customer subscribes or not (\$59.49 vs. \$59.87). Subscribers are simply a smaller group (27% of customers) — the subscription program is currently a segmentation tag, not a spending multiplier.

Loyalty & Fulfillment Patterns

REPEAT BUYERS BY SUBSCRIPTION STATUS

Subscribed	Repeat Buyers	Rate
No	2,518	88.4% of 2,847
Yes	958	91.0% of 1,053

Subscribers do come back slightly more often (91.0% repeat rate vs. 88.4%), but the gap is small. Combined with flat average spend, subscription status is a weak lever on its own — its main value is a marginally higher retention rate, not bigger baskets.

AVERAGE SPEND BY SHIPPING TYPE

Shipping Type	Avg. Purchase
Express	\$60.48
Standard	\$58.46

Customers who choose Express shipping spend about **3.5% more** per order on average than Standard-shipping customers — a small but consistent signal that faster fulfillment correlates with slightly higher basket value.

REVENUE BY AGE GROUP

Age Group	Revenue
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Notably flat: the gap between the highest-revenue age group (Young Adult, \$62.1K) and the lowest (Senior, \$55.8K) is under 11% — age is not a strong differentiator here, unlike gender or customer segment.

What Sells, What Rates Well

TOP-RATED PRODUCTS

ITEM	AVG. RATING
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

MOST HEAVILY DISCOUNTED

ITEM	DISCOUNT RATE
Hat	50%
Sneakers	49%
Coat	49%
Sweater	48%
Pants	47%

Discounting isn't propping up weak ratings

Hat appears on both lists — it carries the deepest average discount (50%) yet still lands in the top-5 rated items (3.80). Ratings across the whole catalog stay tightly banded (3.78–3.86), so heavy discounting looks like a pricing/promo strategy rather than a reaction to poor product quality.

TOP ITEMS BY ORDERS, PER CATEGORY

CATEGORY	RANK	ITEM	ORDERS
Accessories	1	Jewelry	171
Accessories	2	Sunglasses	161
Clothing	1	Blouse	171
Clothing	2	Pants	171
Footwear	1	Sandals	160
Outerwear	1	Jacket	163
Outerwear	2	Coat	161

Order counts for category leaders sit in a tight 145–171 band — no single product dominates its category the way Male customers or the Loyal segment dominate their dimensions. Demand is broad-based across the catalog.

Interactive BI Dashboard

All findings above are consolidated into a single filterable dashboard — sliced live by subscription status, gender, category and shipping type — so the underlying patterns can be explored rather than just read off a table.

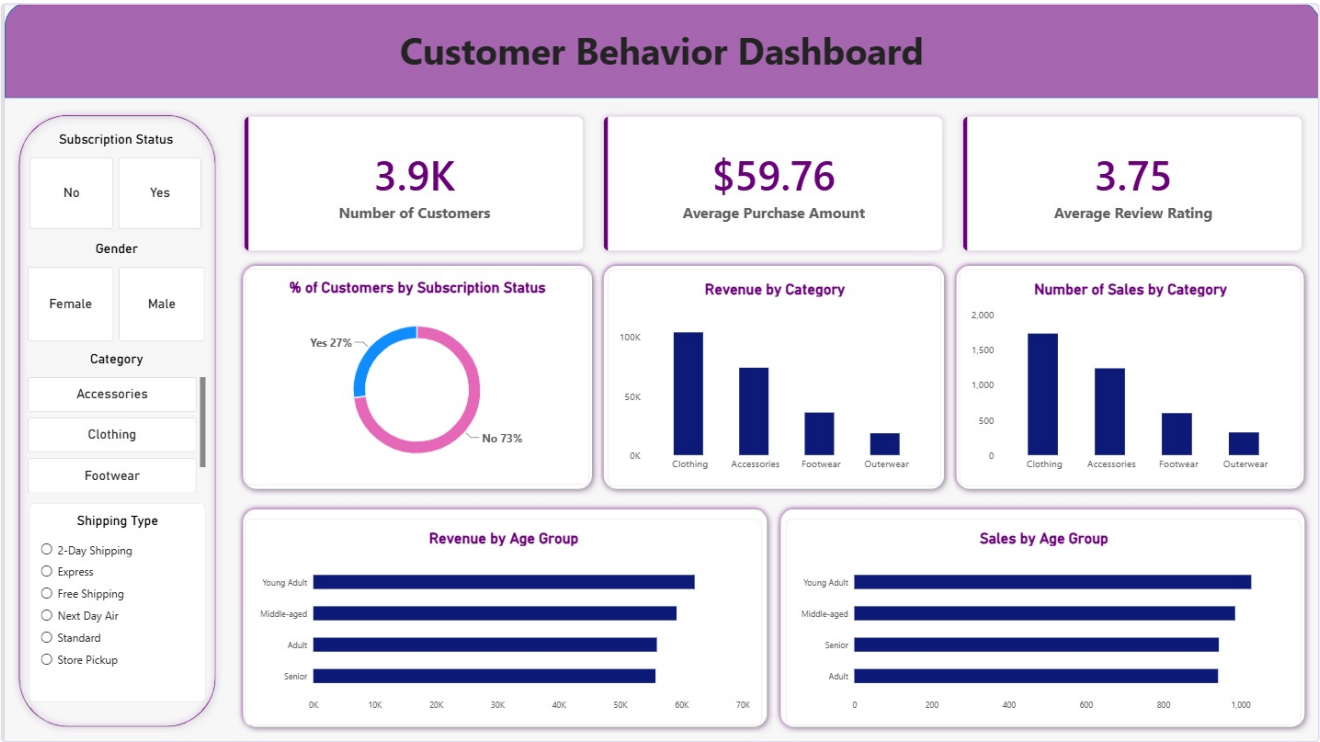


Fig. 1 — Customer Behavior Dashboard: KPI summary, subscription split, revenue & sales by category, and revenue & sales by age group.

Insights & Next Actions

1. Investigate the gender revenue gap before acting on it

Male customers drive 67.7% of revenue. Before designing campaigns around this, check whether it's driven by basket size, order frequency, or category mix (e.g. higher-ticket categories skewing male) — the current data doesn't yet say why.

2. Redesign the subscription program around a real incentive

Subscribers spend the same and only churn slightly less than non-subscribers. If the goal is to grow subscription revenue rather than just subscription count, the program needs a spend-linked benefit (tiered rewards, free Express shipping) rather than status alone.

3. Use Express shipping as a soft upsell signal

The \$2 average gap between Express and Standard spenders is small but consistent — worth testing whether offering free Express above a spend threshold increases basket size further.

4. Re-evaluate deep discounting on already well-rated items

Hat, Sneakers and Coat carry ~48–50% discounts while rating comparably to full-priced items. Given ratings don't justify the depth of discount, these may be over-discounted relative to demand.

5. Treat age and category demand as stable, not a growth lever

Both are evenly distributed with no major outliers — better spent optimizing operations (inventory, fulfillment) around this evenness than chasing an age or category "winner" that doesn't clearly exist in the data.

TOOLS & METHOD

Python (pandas)

SQL (aggregation, ranking, joins)

PostgreSQL · SQLAlchemy

BI Dashboard / data viz

Customer segmentation